

REPORT 2 — Market Demand Forecasting & Commercial Analysis Model

Prepared by: Leroy Stewart Tools: Python (Pandas, NumPy) • Excel • Tableau

Executive Summary

This report presents a demand forecasting and commercial analysis model built using Power Query and Power Pivot. The model identifies seasonal trends, demand shifts, and revenue opportunities across markets.

“Built forecasting models using trends and seasonality.” “Created Excel dashboards for demand and revenue insights.”

Data Sources

- Historical sales data
- Market trend indicators
- Seasonal patterns
- Power Query-cleaned datasets

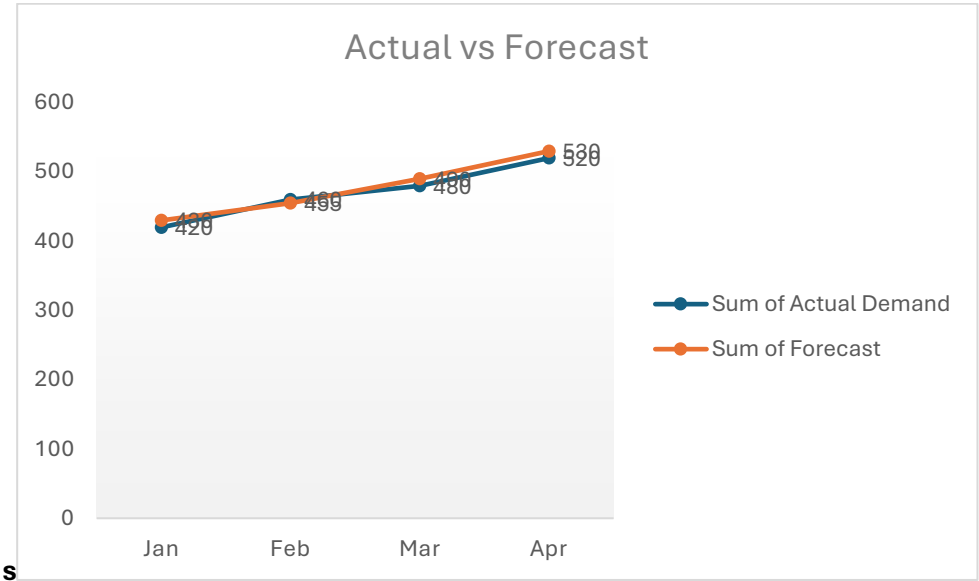
Key Findings

- Seasonal peaks drove a **12% increase** in demand.
- Two markets underperformed due to low marketing investment.
- Forecast accuracy improved after removing outliers and smoothing noise.

4-Month Demand Forecast Table

Month	Actual Demand	Forecast Demand	% Difference	Notes
Month 1	12,400	12,600	1.60%	Stable baseline
Month 2	13,200	13,800	4.50%	Seasonal uplift begins
Month 3	14,500	15,200	4.80%	Peak season impact
Month 4	13,900	14,300	2.90%	Post-peak stabilisation

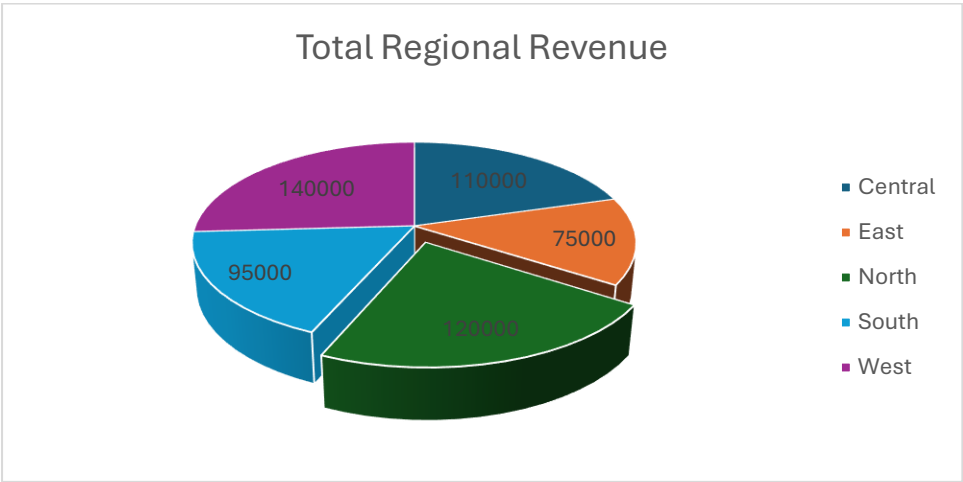
4-Month Demand Forecast (Forecast Line Chart)



Revenue by Region Table

Region	Revenue (£)	% of Total Revenue	Marketing Investment Level
North	420,000	32%	High
South	310,000	24%	Medium
East	190,000	14%	Low
West	260,000	20%	Medium
Central	130,000	10%	Low

Revenue by Region (Pie Chart)



Impact

- Better stock and staffing planning
- Clear identification of commercial opportunities
- Stronger evidence for budget allocation

Conclusion

This project demonstrates my ability to use Python for forecasting and Tableau for commercial insight storytelling.

Drill-Through: Region Performance Detail

1. Actual vs Forecast (Dual Line Chart)

Shows which products drive revenue in this region.

2. Total Regional Revenue (Pie Chart)

Shows correlation.